INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II Fourth Semester Examination, March- 2025

Subject Code: BCAAE401

Subject Name: Software Engineering

Full Marks: 30 Set No: 1 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What are the different categories of software development projects according to the COCOMO estimation model? 2

2. What is the major shortcoming of basic cocomo model? 2

3. Why UML is used for modelling? 2

4. What is meant by code inspection? 2

5. Define context diagram. 2

6. What are the driver and stub modules in the context of unit testing? 2

Part - B

Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Suppose that a project was estimated to be 400 KLOC. Calculate the effort and development time for each of the three modes i.e., organic, semidetached and embedded. 5

8. What are cost drivers and what is the purpose of using them? 5

9. Explain the conceptual model of UML. 5

Part - C

Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Describe different coding guidelines. Also, differentiate between verification and validation. 10

11. Draw the RMS calculating software of DFD level 0,1, and 2. 10

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , Fourth Semester Examination, March- 2025

Subject Code: BCAAE401

Subject Name: Software Engineering

Full Marks: 30 Set No: 2 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What is the role of a software architect? 2

2. What are the disadvantages of basic cocomo model? 2

3. What is a Data Flow Diagram (DFD)? 2

4. What are the benefits of using Object-Oriented Programming (OOP) in software engineering? 2

5. Explain the concept of 'reusability' in software engineering. 2

6. What are the different types of UML diagrams? 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Consider a project was estimated to be 1000 KLOC. Calculate the effort, development time and manpower for each of the three modes i.e., organic, semidetached and embedded. 5

8. What is the difference between 'white-box' and 'black-box' testing? 5

9. Explain the class diagram in UML. 5

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Explain the unit testing, its objectives and the techniques used. 10

11. Design a DFD for Library management system up to label 2. 10

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Fourth Semester Examination, March - 2025

Subject Code: BCAAE401

Subject Name: Software Engineering

Full Marks: 30 Set No: 3 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What is the function of a 'test case'? 2
2. What are the basic principles of 'object-oriented design'? 2
3. What is a process? 2
4. What is 'load testing'? 2
5. What is the significance of 'user acceptance testing'? 2
6. What is data dictionary? Is there any use of data dictionary in Data Flow Diagram (DFD) justify your answer? 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. What do you understand by the term functional independence in the context of software design? What are the advantages of functional independence? How can functional independence in a software design be achieved? 5
8. What is meant by the term cohesion in the context of software design? Is it true that in a good design, the modules should have low cohesion? Why? 5
9. Is system testing of Object Oriented Programing any different than procedural programing? Justify your answer. 5

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. What do you mean by modular design? What are the different factors that affect the modularity of a design? How can you assess the modularity of a design? What are the advantages of a modular design? 10
11. What do you understand by the term integration testing? Which type of defects are uncovered during integration testing? What are the different types of integration testing methods that can be used to carry out integration testing of a large software products? 10